

Embedded Systems Engineer with 2+ years building production embedded electronics end to end — from concept and hardware architecture through schematic capture, PCB layout and Embedded C/C++ firmware on STM32 and ESP32, to board bring-up and on-site commissioning. Work spans dual-fuel (LPG/CNG) engine control units, injector driver electronics and industrial monitoring systems delivered to manufacturers including Delphi TVS, Indo Tech Transformers and FI Auto Components.

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SKILLS

Firmware

- Embedded C / C++ · interrupt-driven · state machines · drivers: GPIO, ADC, PWM, timers
- PlatformIO · STM32CubeIDE · Arduino IDE · Git
- Learning: FreeRTOS · STM32 bare-metal · Linux (Debian)

Microcontrollers

- STM32 F1/F4 · ESP32 / S3 · ESP8266 · Arduino Mega/UNO
- ATtiny 3216 / 2313 — bare-IC design, UPDI

Engine & ECU Electronics

- Dual-fuel ECU (LPG/CNG) · injector drivers · fuel control logic
- Engine sensing: MAP, EGT, RPM · sensor calibration · CAN interfacing

Hardware & PCB

- Schematic capture · PCB layout · BOM · bring-up · validation
- MOSFET driver circuits · linear supplies · buck converters · flyback and reverse-polarity protection
- Altium Designer · KiCad · EasyEDA · Proteus

Protocols & Data

- CAN · RS485 · UART · SPI · I2C · LoRa · HTTP/JSON · SD/SPIFFS logging

Test & Debug

- Oscilloscope · multimeter · bench power supply
- ST-Link SWD · UPDI programming · USB-UART
- Custom UART capture tool · board-level fault analysis

WORK EXPERIENCE

Embedded Product Engineer
Spark Invotech Pvt Ltd, Chennai

2024 – Present | 2+ years | DPIIT-recognized embedded systems company I co-founded

- Architected and delivered commercial embedded products end-to-end — requirements, hardware architecture, schematic capture, PCB layout, Embedded C/C++ firmware, board bring-up, validation and customer deployment.
- Led dual-fuel ECU and injector-driver development for IC engines, targeting marine engines and generator sets.
- Commissioned industrial monitoring electronics for Delphi TVS (diesel fuel-injection systems manufacturer) and Indo Tech Transformers Ltd (power equipment manufacturer), including on-site validation and technical documentation.
- Engineered products across industrial automation, automotive electronics (ECU / OCU), industrial IoT monitoring and agritech; worked directly with client engineering teams from requirement definition through on-site validation and handover.

R&D Engineer Intern
AbhiVijay Technologies, Chennai

Jan 2024 – Apr 2024

- Developed ECU and OCU hardware — R&D, hardware design, testing and system-integration validation for automotive electronic control units.

KEY PROJECTS

Cold Chamber Data Logging System
Delphi TVS · delivered

- Architected a 3-node distributed data logger that records true temperature saturation inside a cold-start test chamber, so the -20 °C soak period is measured rather than assumed before engine cold-start testing.
 - Engineered a command/ACK protocol with 3-retry logic, dual-mode operation (live transmit / silent store) and session persistence across power cycles; validated on-site before customer handover.
- Tech: ESP32, RS485 half-duplex (DE/RE), LoRa E32 (~200 m as deployed), SPIFFS flash logging, Python PC interface

EDUCATION

M.E. — Embedded System Technologies

Sri Sai Ram Engineering College,
Chennai
CGPA 7.82

Completed while leading commercial embedded product development at Spark Invotech Pvt Ltd

Modules: Embedded Systems Design · ARM Cortex Architecture · Real-Time Systems (FreeRTOS) · Advanced Microcontrollers (STM32) · Embedded Linux · DSP

B.E. — Electrical & Electronics Engineering

Sri Sai Ram Engineering College,
Chennai
75%, First Class

Modules: Microprocessors & Microcontrollers · Power Electronics · Control Systems · Analog & Digital Electronics · Sensors & Transducers

AWARDS

Youth Co:Lab (UNDP) — shortlisted Top-20 nationally, finalist; five-day intensive workshop

Smart India Hackathon 2025 — team mentor

DPIIT Startup Recognition and IVP Voucher Grant — Spark Invotech Pvt Ltd

Coimbatore Startup Summit, Oct 2025

Incubation-lab hands-on projects, 2022 – 2024

CERTIFICATIONS

IIT Bombay — Arduino

IIRS / ISRO — 5 certifications

IEEE Learning — 3 certificates

NPTEL — 2 certifications

LANGUAGES

English — Full Professional

Tamil — Native

KEY PROJECTS (CONTINUED)

Compressor Health & Power Monitoring System Indo Tech Transformers Ltd · POC completed

- Developed an **industrial IoT logger for compressor health and power monitoring** — temperature, filter condition, pressure, current and voltage — with RTC timestamping, local storage and cloud sync.
- Integrated instrumentation-grade sensors with signal conditioning and calibration, plus a P10 LED display for on-site readout.

Tech: ESP, PT100, ZMPT101B (voltage), SCT013 (current), RTC, SD logging, HTTP/JSON cloud APIs, P10 LED display

Production Quantity Monitoring System FI Auto Components Pvt Ltd · POC completed

- Built a shop-floor **machine output monitoring system** that counts produced components and pushes live production data to a dashboard; POC completed and under evaluation for plant-wide rollout.

Tech: ESP module with microcontroller, sensor input, HTTP/JSON dashboard integration

Dual-Fuel Engine Control Unit (LPG/CNG) Indigenous low-cost ECU — Spark Invotech

- Developed an indigenous, low-cost, customizable ECU for converting IC engines to **dual-fuel LPG/CNG** operation, targeted at marine engines and generator sets.
- Integrated **real-time engine sensing** — MAP, exhaust gas temperature and engine speed via precision ADC — with in-house calibration, and implemented the fuel and injector control logic across multiple firmware versions.

Tech: ATtiny 3216 (final build), ATmega / Arduino and ESP32 (pilot versions), ADS1115, MAP/EGT/RPM sensors, CAN module, Embedded C, PlatformIO

Injector Multiplier ECU channel-expansion module

- Designed a power-electronics module multiplying 4 ECU injector channels to drive up to **16 injectors**, letting large multi-cylinder engines run on a low-channel, low-cost ECU.
- Implemented **12 V / 8 A peak-per-channel driver stages** with flyback and reverse-polarity protection on a custom PCB.

Tech: MOSFET switching, protection circuitry, custom PCB, power-stage layout

Oil Control Unit (OCU) RPM-based oil regulation — LPG-converted marine engines

- Designed, manufactured and verified **two hardware versions on custom PCBs** that regulate oil delivery from live engine RPM, protecting marine engines converted to LPG where full gas conversion raises running temperature and risks seizure.

Tech: ESP8266, PWM control, MOSFET/TIP122 power stage, protection circuitry, custom PCB

Acoustic Vehicle Alerting System (AVAS) for EVs Electric vehicle pedestrian-safety prototype

- Prototyped a **speed-dependent acoustic warning system** on ESP32-S3 where alert tone and volume rise in proportion to vehicle speed, extending engine-sensing work into EV safety hardware.

Tech: ESP32-S3, PWM audio generation, analog speed input, Embedded C/C++, PlatformIO

Additional Projects

Trust-funded design · incubation lab, 2022 – 2024

- Portable Biogas Plant** — automated stirring with water-scrubbing and MDEA gas purification, plus mobile monitoring and control.
- Cloud Fusion Logger** — smart IoT data logger and controller with predictive-maintenance features.
- Mars Rover** — built the **science subsystem** with full automation of the sample and science-test sequencing, integrated with the rover's multi-sensor and motion control.